

## 5-Mark Questions & Answers -- DBMS Module 5: Query Optimization & Transaction Concurrency Control

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**Q1. Explain cost estimation for selection and join operations in query optimization.**

1. Selection Cost ( $b_r$  = blocks of  $r$ ):

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**Q2. Explain 2-Phase Locking Protocol (2PL) and Multiple Granularity Locking (IS, IX, SIX).**

- \* 2PL: Growing Phase (Acquire locks) -> Lock Point -> Shrinking Phase (Release locks). Guarantees conflict serializability.

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**Q3. Explain Timestamp Ordering Protocol and Thomas Write Rule.**

- \* Timestamp Ordering: Assigns  $TS(T_i)$ . Checks  $TS(T_i) < W-TS(X)$  for Read and  $TS(T_i) < R-TS(X)$  for Write. Aborts if violation occurs (Deadlock Free).

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**Q4. Explain Log-Based Recovery (Deferred vs Immediate update) and Checkpointing.**

- \* Deferred Update (NO-UNDO/REDO): Writes postponed until Commit. Requires only REDO during crash recovery.

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**Q5. Describe Two-Phase Commit (2PC) protocol for distributed transactions and TCL commands.**

- \* 2PC Protocol:

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